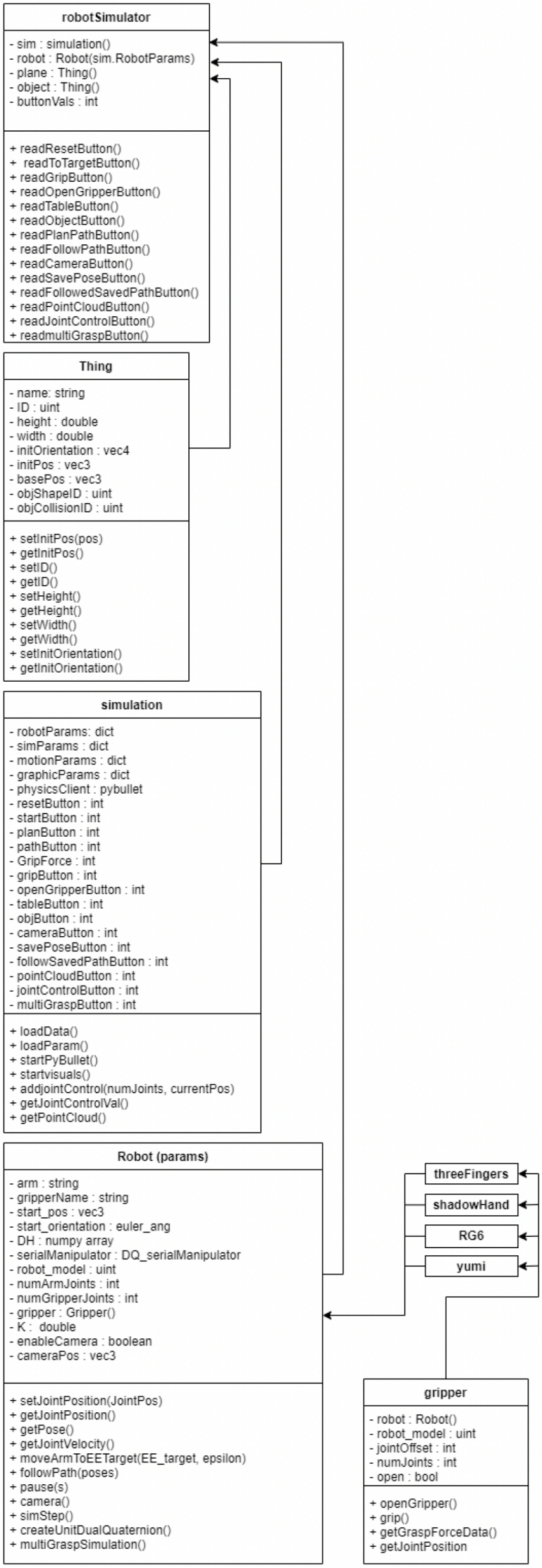
robotSimulator
- sim : simulation()
- robot : Robot(sim.RobotParams)
- plane : Thing()
- object : Thing()
- buttonVals : int
+ readResetButton()
+ readToTargetButton()
+ readGripButton()
+ readOpenGripperButton()
+ readTableButton()
+ readObjectButton()
+ readPlanPathButton()
+ readFollowPathButton()
+ readCameraButton()
+ readSavePoseButton()
+ readFollowedSavedPathButton()
+ readPointCloudButton()
+ readJointControlButton()
+ readmultiGraspButton()
Thing
- name: string
- ID : uint
- height : double
- width : double
- initOrientation : vec4
- initPos : vec3
- basePos : vec3
- objShapeID : uint
- objCollisionID : uint
+ setInitPos(pos)
+ getInitPos()
+ setID()
+ getID()
+ setHeight()
+ getHeight()
+ setWidth()
+ getWidth()
+ setInitOrientation()
+ getInitOrientation()
simulation
- robotParams: dict
- simParams : dict
- motionParams : dict
- graphicParams : dict
- physicsClient : pybullet
- resetButton : int
- startButton : int
- planButton : int
- pathButton : int
- GripForce : int
- gripButton : int
- openGripperButton : int
- tableButton : int
- objButton : int
- cameraButton : int
- savePoseButton : int
- followSavedPathButton : int
- pointCloudButton : int
- jointControlButton : int
- multiGraspButton : int
+ loadData()
+ loadParam()
+ startPyBullet()
+ startvisuals()
+ addjointControl(numJoints, currentPos)
+ getJointControlVal()
+ getPointCloud()
Robot (params)
- arm : string
- gripperName : string
- start_pos : vec3
- start_orientation : euler_ang
- DH : numpy array
- serialManipulator : DQ_serialManipulator
- robot_model : uint
- numArmJoints : int
- numGripperJoints : int
- gripper : Gripper()
- K : double
- enableCamera : boolean
- cameraPos : vec3
+ setJointPosition(JointPos)
+ getJointPosition()
+ getPose()
+ getJointVelocity()
+ moveArmToEETarget(EE_target, epsilon)
+ followPath(poses)
+ pause(s)
+ camera()
+ simStep()
+ createUnitDualQuaternion()
+ multiGraspSimulation()
threeFingers
shadowHand
RG6
yumi
gripper
- robot : Robot()
- robot_model : uint
- jointOffset : int
- numJoints : int
- open : bool
+ openGripper()
+ grip()
+ getGraspForceData()
+ getJointPosition